

Status Report — 16 September 2026

Branch `main-2026` at `c92bf36`, 16 Sep evening · 5,957 tests passing, 11 skipped · prepared for James Gray; shareable with Tony Ford and Howard Thurley

This page is the engine-level detail for the estimators. **The company-wide picture — this workstream plus Yogesh Kumar's and Muhammad Yazir's — is the 16 September set:** the programme status report (issue 3), the covering e-mail to the MD and FD, and the portal, all listed at the foot of this page.

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Where each job stands

Howard — 7332-01 RERUN ASSESSED

All nine points of Howard's reply are implemented, plus both rerun blockers: the laser set-up shares only across a **proven** one-sheet nest, and the plater pack times are reconciled per unit. Plating and freight are shown as **explicitly awaiting current quotes** — no figure off a manual estimate is retained anywhere. The 17:36 rerun is assessed; the five laser rows, the stated weld and dress times and the two pack rows all landed.

Open from that book: the shared set-up on the two nested parts, the brushing minutes and pack minutes on the frame, a tube-bend flag still on the sawn leg, and the plating specification not identified on the run. The gauge cause was found and fixed from the book alone; the rest need the run's own `output/json/7332-01.json` from the laptop.

Howard — 10975-02 EMAIL HELD

Figures stable since the 14:28 book (Description, Rev B, four quantity breaks, 1-off deliberate) — the engine's own output, not anybody's sheet, and they move when the build moves. Needs one clean rerun on the current build before the covering e-mail, which will state that the 1-unit run is deliberate and answer his points.

Tony — 11908-21 + 5 new

NEXT IN QUEUE

Tim — 11350-02 / -01

ANSWERED, 2 FIGURES OUT

11908-21's joinery route is scoped correctly (bench rate now reaches only faced board). Still open before rerun: MC J emission proof on the book, edge-banding metres (his input), box-and-pallet method into config, and a guard so a **purchased, customer-supplied or pre-assembled board parent** is not given bench work merely for having child records. The five new jobs run as controlled learning passes, findings fed back per job.

Four points, two of them about where a figure came from and two about why a line carried none. Both "where from" questions are answered by arithmetic that reproduces the booked row — pemming at 15 s a pressed fastener (his own 12120 sheet), powder by weight and never by RAL. Both "why none" cases are fixed generically (below). **Out to him:** the powder rate per kg, the coverage figure, and whether 15 s still holds. 11350-01 is the same job at another size and re-runs beside it.

The fortnight in numbers

Counted rather than characterised. The last two are the honest ones: both are gaps, and both have an owner.

11

jobs run in two weeks, against a minimum of five a week

3 / 3

estimators have returned a review — six job-reviews between them

~40

points raised by the estimators, and every one is now a rule, not a patch

69

numbered decisions on the register, D-016 to D-082, each with its source and its proof

5,957

tests passing across 381 files — they execute the engine, they do not read it

272

commits since the 7 September report

0 / 3

supplier price lists loaded — Elite, Eagle, Thermaset. Every unpriced bought line traces back here

0 / 11

parity comparisons filed for this fortnight's jobs — see below

Parity — what we can prove, and what we cannot

What it is. The engine's workbook read against the estimator's own completed sheet, block by block. Both sides are reconciled to their own totals before anything is compared, differences are signed in both directions, and the inputs behind a money gap — a blank, a gauge, a quantity, a nest — are reported beside it. It files a ledger row per job, so the picture builds rather than being re-argued each time.

When it runs. From the portal against a finished pair, or at run time where the estimator attaches the manual sheet up front.

Where it stands on the eleven: no cell-level comparison is filed for any of them, because no manual sheet has been attached to a run. The ledger in source control holds one row, from a July job. Saying otherwise would be the one thing this programme does not do — the proof is the log, not the theory.

So the parity evidence for this fortnight is the estimators' own verdicts rather than a computed variance, and there are two of them:

- **Tim on 11350-02:** *"Costs come out really close."* His words, and the closest thing to a parity verdict on a current job.
- **12349-02:** the three defects his review found were fixed the same day and every line was then reconciled to his sheet. It is held as a replay gate, so no later change can quietly move it.
- **The other nine:** no verdict either way. That is not a claim of accuracy and should not be read as one.

What turns this into a table. One command a job on the laptop, against the manual sheet wherever one exists:

```
python src/parity_run.py --ai output/json/<job>.json --manual <manual sheet>
```

Each run files a ledger row, and the eleven stop being an impression. Where a job was estimated by the engine first, the comparison can be added when the manual sheet arrives.

Why no figure off a manual sheet appears here. By rule, a number from a manual estimate is not retained in our register, documentation or reports. Parity is therefore reported as a variance and a direction, never by printing the estimator's own amount.

What is live and generic

RULE	WHAT IT DOES	PROOF

One laser row per component	Each component gets its own LASM row and rate — no blended rate across a strap and a cap. Rows of one material and gauge share ONE department set-up, allocated only when the engine's own nesting proves the whole order fits one sheet; unproven nests keep full set-ups and say so.	D-066/074 · executed tests: five components never create five set-ups
0.9 mm steel → 1.0 mm	A confirmed production-substitution rule (data, not code): costed at the gauge the shop buys, drawn figure kept on the line, estimator-confirmed thickness stands it down. Mild steel and Zintec only. A confirmed rule of this kind cannot be outvoted by the drawing readers — the quorum that kept one part at 0.9 is now barred from overruling a ruling.	D-068/078 · all three states executed
Mitred tube is sawn, not bent	The tube-bender needs a round/oval section and a radius. A square leg whose only evidence is 45° callouts loses the op, citing Howard's ruling; measured bends are never second-guessed.	D-069
No default acrylic peel	"Nothing fixed for this" — the standing MANA allowance comes off unless the drawing states the work; the absence is flagged with how to reverse it.	D-070
Customer finish standards	"M&S dress all seen welds; TTI none" — kept as a per-customer rule with Howard's name on it. Governs the engine's inference only; a drawing stating dressing is never overruled.	D-071
Two packs, two rows	A plated part packs out (its own PACM row) and packs back. The	D-072/075

total is unchanged; each figure is now checkable on its own line.

Board assembly bench line	Every multi-part board assembly gets a bench-fitting line — the BOM's own structure says so. The <i>rate</i> is scoped (below); the <i>rule</i> is generic.	D-067
Faced board keeps its facing	MF MDF stated on a title block survives the canonical collapse to MDF, is promotion evidence on its own, and prices by the purchased sheet — closing the 4× plain-core under-charge path.	D-067
One workbook, every quantity	An enquiry asked at 10, 50, 250 and 1,000 produces one spreadsheet , not one per quantity. The estimators' own Material Price Break tab is filled by the engine and a Quantity Breaks tab recalculates the whole estimate at each quantity, with order-level freight divided by that quantity rather than landed on every unit. The base sheet is deliberately run at one unit, where each department's set-up is visible instead of hidden in a volume average.	"Still generating multiple s/sheets" — James, 15 Sep. The sweep still computes the figures; it no longer files copies

An unpriced line names whose gap it is	Three lines on Tim's comms bar said only "needs a rate". Two carried a code the engine minted from the description (BI-NUT, BI-PEMSTUD) because the pack gives none — nothing can be looked up against a code we wrote, and it sits in the code column looking like a part number. The third carried DBR60, a real code that was asked and found nothing. Each now says which, because one is closed by a code on the pack and the other by loading the catalogue.	D-079/080 · executed tests: a code we invented is not a code Tim can look up
A figure states the sum that made it	Pemming is the MANM department rate × 15 s a pressed fastener (4 clinch at 60/hr and 2 PEM at 120/hr are both 240 inserts an hour). Powder is coated area × coverage × £/kg, with the colour never entering the sum. Both reproduce the booked row exactly — the test a figure passes before it stands.	D-081/082
An unpriced line clears the cell	A line awaiting a price writes an empty cell rather than leaving the template's own formula in place — which is what put #REF! on the Harrods book. No line can look priced because a template remembered a number.	Executed test on a real worksheet

What is scoped pilot knowledge — measured once, not shop constants

FIGURE	MEASURED ON	SCOPE
Joinery 12 / 12 / 12 / 2 / 20 per hr	Tony, 11908-21 (one faced-board tray, 50 off)	CNCJ/EDGE/MC J/BENC/PACJ run rates. Bench 2/hr reaches faced/laminated board only ; plain

		MDF, ply and timber take the labelled house allowance.
Brushing 40 min/unit	Howard, 7332-01 ("similar size units")	Pre-plating brush — method generic, timing size-sensitive, confirm outside similar size.
Weld 30 / dress 20 min	Welding dept via Howard, 7332-01	Per-joint rates derived from the measured joint count; stated times always beat medians.
Pemming 15 s per pressed fastener	Tim, 12120 sheet (clinch ×4 @60/hr, pem ×2 @120/hr)	Both of his rows state 240 inserts an hour. Applied wherever the reconciled BOM holds self-clinch or PEM hardware; out to him to confirm, and to say whether a clinch nut and a PEM stud should differ.
Set-up model (shop-wide)	Confirmed on Tony's joinery AND Howard's metal sheet	Hours = department set-up once per order + run × qty. The <i>model</i> is shop-wide; the rates above stay pilots.

Each row carries the job and the person it was measured on, which is what makes a change in it visible: when the same operation is measured again on a second job the two figures sit side by side, and the difference is a question for the department rather than a silent overwrite. A pilot becomes generic only when a second independent job or the department itself confirms it.

What must never inherit

Historic estimates teach; they do not supply prices. *"A number copied from an estimator's sheet is not a price source, even as a reference." "A rate taken from a single estimate is a scoped pilot, never generic, until a second independent job or the department itself confirms it."*

- **Harrods 01 plating** — the quoted figure is retained nowhere. What it taught lives on as a rule: Harrods 01 is decorative plating that needs a fresh, identified plater quote for each job; until one exists the line is not priced.
- **Plater freight** — the reference figure from Howard's review is retained nowhere; a plated route still carries a freight line, raised **awaiting a current transport quote** (owner: SDI transport) on every job, 7332-01 included. The route inherits; the money does not.

- **Edging rate and tape roll price** — off the estimators' sheets, so no longer held anywhere: the edging specification, the roll length and the length-arithmetic carry forward; the money comes from SDI Live or a supplier source or the line is explicitly awaiting price.
- **Held to by test, not by intent** — the register, the price file, the documentation, the reports, the prompts and the audit payload are all empty of these figures, and a test asserts that a new run cannot discover or display a historic manual-estimate amount anywhere. Mechanism tests use invented figures so they prove the arithmetic without carrying a real one.

Questions owned by people

OWNER	QUESTION
James Gray	Three configured rates have no live source — the board sheet price, the zinc plating rate per kg and the section-stock rate per kg sit in configuration with no SDI Live, catalogue or quote behind them. Under D-078 they are either sourced or turned into awaiting-price lines with an owner.
Tim	Powder, two figures. Is £4.00/kg still right for standard powder? And the coverage — we hold 0.20 kg/m ² , the workbook template's own calculator holds 0.1667. They disagree by a fifth. His proposal to put the powder reference on the drawing is accepted in principle and retires both.
Tim	Pemming, 15 s a pressed fastener — still right, and the same for a clinch nut and a PEM stud?
Tim	DBR60 a rate or the catalogue row it should read; and SDI codes for the M4 wing nut and M4×8 PEM stud if they have them.
Howard	Two PACM set-ups on the split pack rows (one per occasion) — or does the bench regard them as one? The rerun book flags it either way.
Howard	Acrylic laser throughput: the corpus figure against his own. Whose holds? His answer goes in the stated-rates register, not code.
Tony	Banded metres on 11908-21 (only visible edges band — his judgement, not a drawing number).
SDI transport	A current haulage quote for any new plated job — the line asks for it by name.
Runner	The 7332-01 run's own JSON, so the five open items on that book can be diagnosed; then clean reruns — 10975-02 for the e-mail, then 11908-21.

Elsewhere in SDI Intelligence

The engine is one of three workstreams. The other two are reported in full in the company-wide set below; this is where they stood on 16 September.

SDI Technical Design Intelligence

P2 HARDENING

Yogesh Kumar. Production Design Extraction hardened after its 2 September approval — the two checkouts untangled, the fifteen missing BOM previews diagnosed, extraction history and a metrics endpoint added, tests from about 240 to 363. Six demo videos with write-ups. P2 keeps 29 September; P3.1 has started, with Ubuntu and Docker installed for the PDM MVP. Watch-points: a day or two absorbed by demos and new design-team requirements, and the server and hardware quotes still to be finalised.

SDI Creative Design Intelligence

DRAWING SEARCH LIVE

Muhammad Yazir. **SDI Drawing Search Intelligence is live** for the design and technical teams, out of UAT and ahead of its 25 September date: 32,422 drawings searched by what is printed on the sheet, with the SolidWorks or 3ds Max file opened from the result. Ian tested it before it went in. **Client Briefing is back on the agenda** after its hold from 4 September — brief intake, with Jonathan approving every brief; two briefs through it so far.

Also shipped today

- **SDI Drawing Search Intelligence** is a menu item on the portal and both estimating pages, directly below its guide, opening the application in its own tab — one change, every deployment (8071 / 8072 / server). It is now described as live rather than in UAT everywhere the site states it.
- Tests across the batch **execute the engine** (`estimate_part()` , the group builder, the allocator on a real worksheet) rather than reading source text — which is how the facing-collapse defect was caught at all.
- Change register rows **D-067** → **D-082**; D-064's scope and D-066's presumption formally superseded, and D-077's register entry withdrawn by D-078 rather than reworded.

The company-wide 16 September set

- **SDI Programme Status, issue 3** — `reports/SDI-Programme-Status-2026-09-16.html` and its PDF. All three workstreams, the shared workstation, and what is open across the programme. Supersedes the 7 September issue.
- **Covering e-mail to the MD and FD** — `docs/SDI_Programme_Update_Email_20260916.html` , with a plain-text twin.
- **The portal** — dashboard, AI Programme, AI Roadmap and the service pages, updated to the same facts on the laptop and the server.

- **The change register** — `docs/CHANGE_REGISTER.md` is the operating contract; every rule on this page carries its row, its source and its proof there.

SDI Intelligence · estimating engine · 16 September 2026 · engine-level detail for the estimators; the programme-level report is issue 3, above.